```
Algo(n)
{
    if n < 5 then return 1
    return Algo(n/2) + Algo(n/2) + Algo(n/2)
}
```

$\boxed{3\,\text{Algo}(n/2)}$

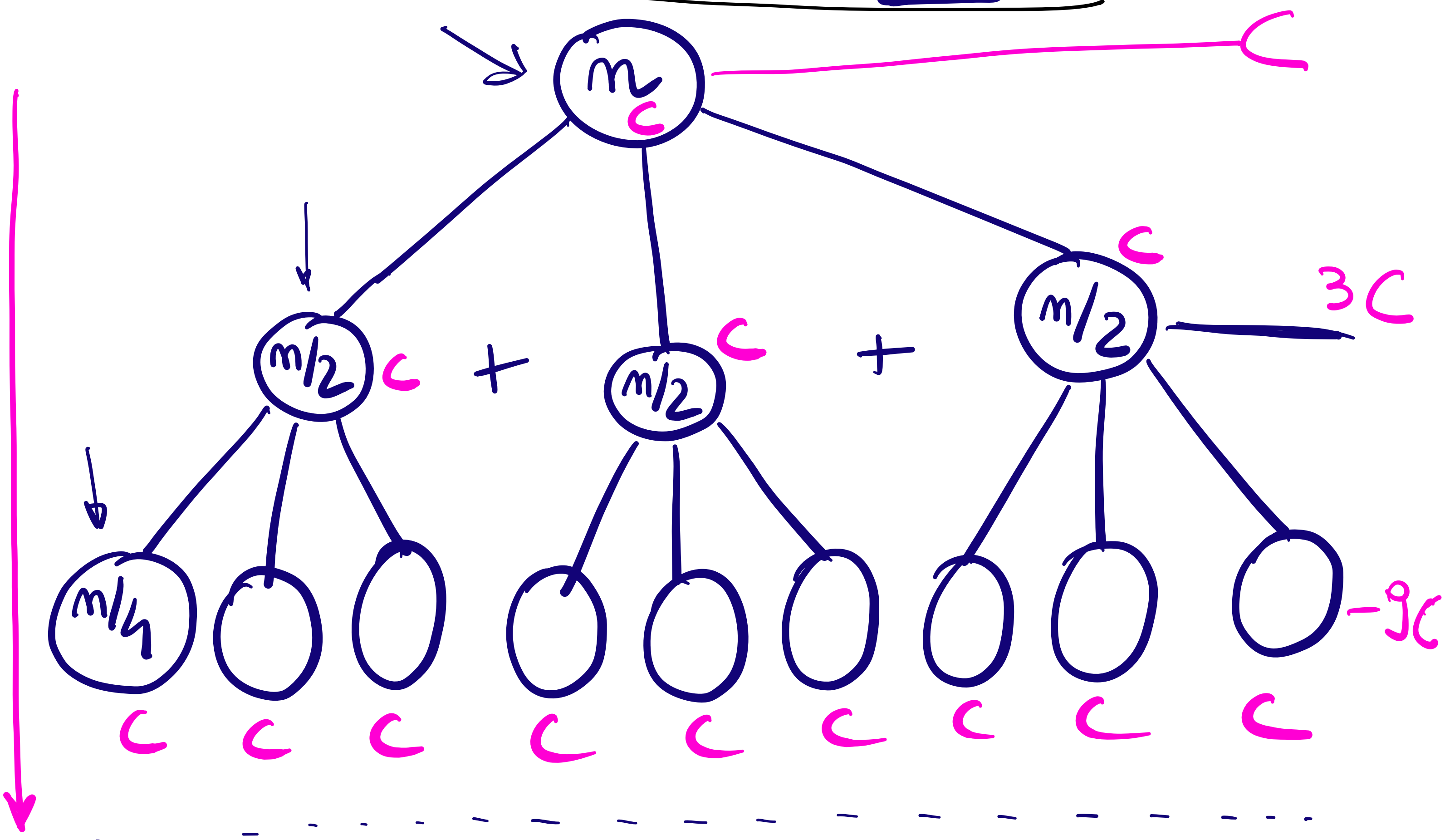
CASO 2 / $\Theta(n^{\log_2 1} \cdot \log n)$

$\Theta(\log n)$

$$T(n) = T(n/2) + C$$

$$n^{\log_2 1} = n^{0} = 1$$

$$T(n) = \begin{cases} c & n < 5 \\ 3T(n/2) + c & n \geq 5 \end{cases}$$



$$\boxed{i = 3^i \cdot c}$$

$$n$$

$$\frac{n}{2}$$

$$\frac{n}{4}$$

$$\frac{n}{8}$$

$$\frac{n}{16}$$

$$\vdots$$

$$\frac{n}{2^i} = 1$$

$$\boxed{i = \log_2 n}$$

$$3^{\log_2 n} \cdot c = n^{\log_2 3} \cdot c$$

$$O\left(n^{\log_2 3}\right)$$

$$T(n) = 3T(n/2) + c$$

METODO DELL'ESPERTO

$$n^{\log_b a} = \underbrace{n^{\log_2 3}} \qquad \underbrace{c}$$

$\boxed{\text{CASO 1}}$ $\quad \Theta\left(n^{\log_2 3}\right)$

$$T(n) = 2T(n/8) + \sqrt[3]{n}$$

$$n^{\log_8 2} \quad // \quad \sqrt[3]{n}$$

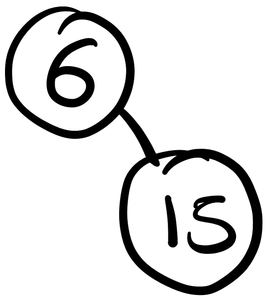
CASO 2

$$\Theta\left(\sqrt[3]{n} \cdot \log n\right)$$

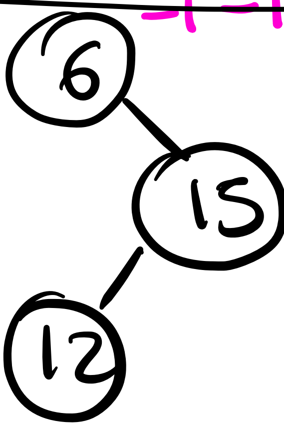
AVL

I: ~~6~~, ~~15~~, ~~12~~, 21, 17, 25, 27, 30

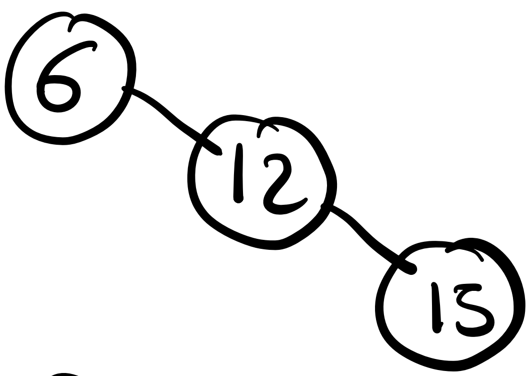
C: 6, 15



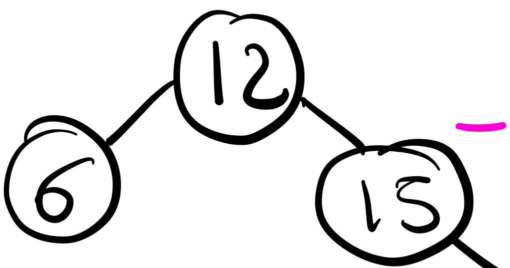
$-1-1=-2$

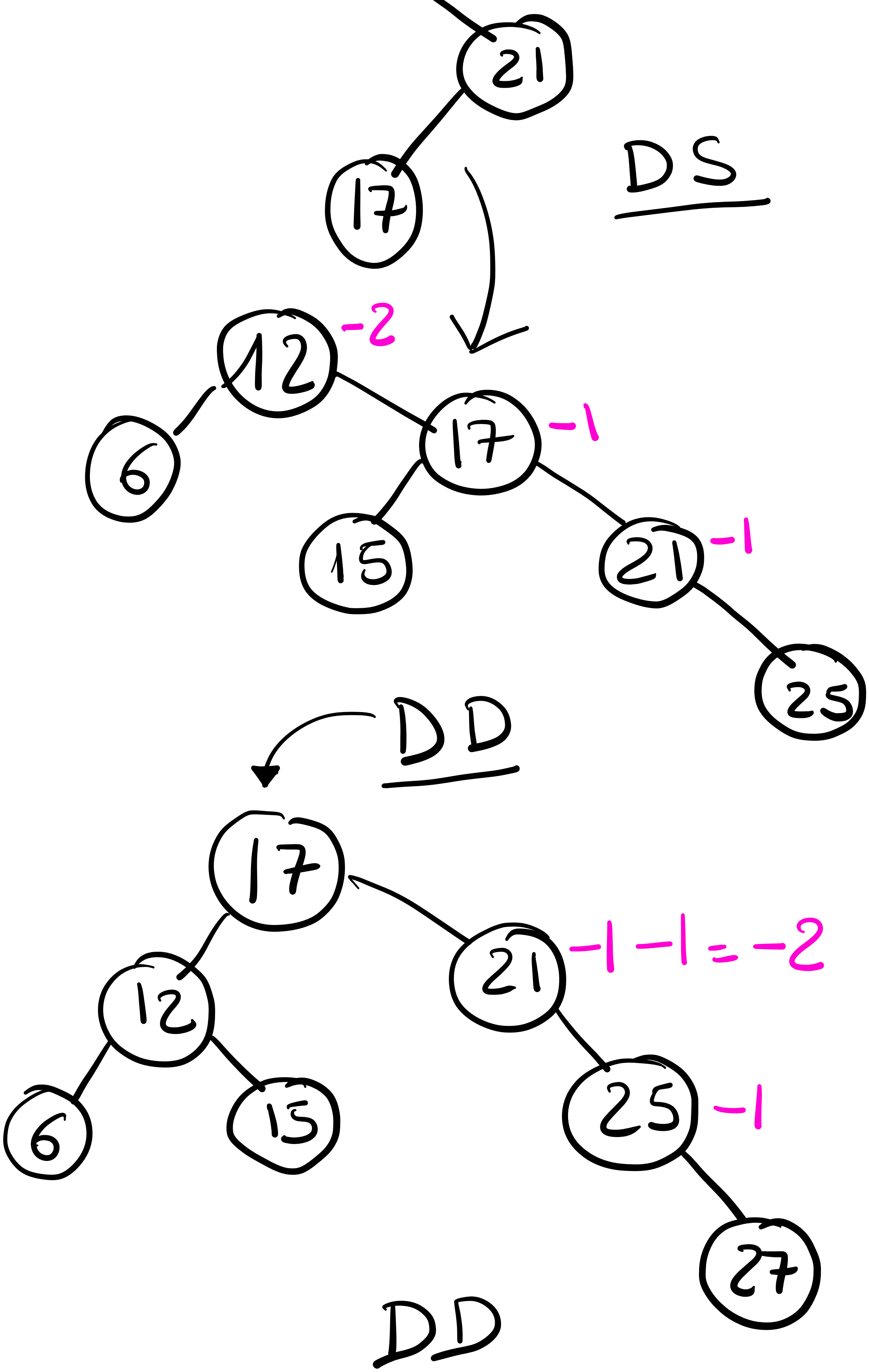


DS

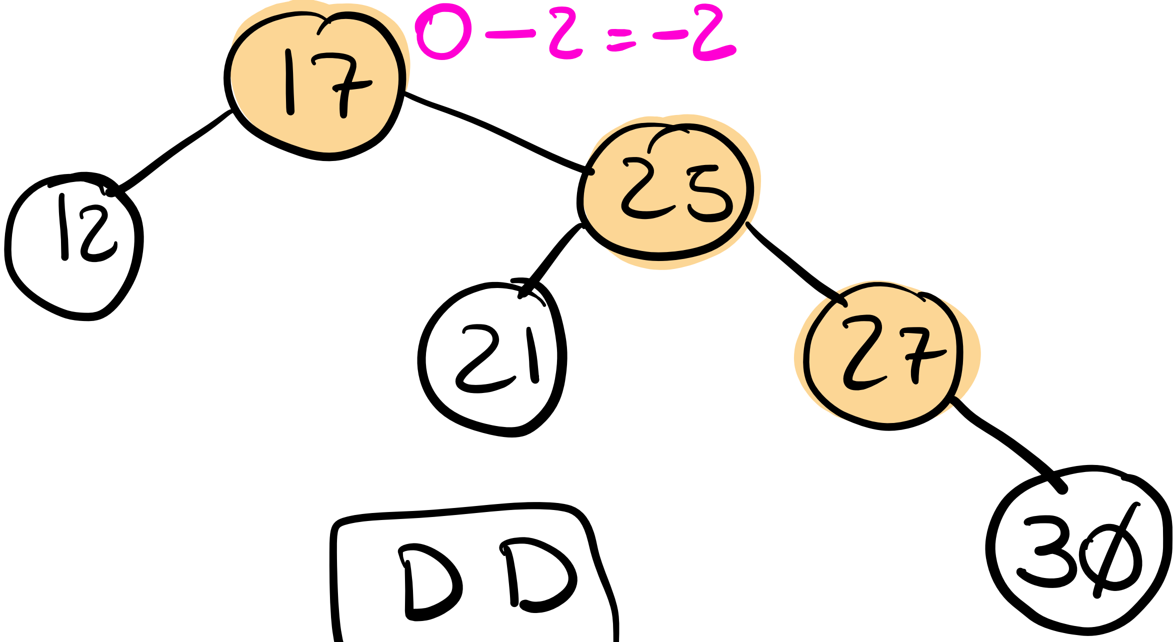


$-1-1=-2$

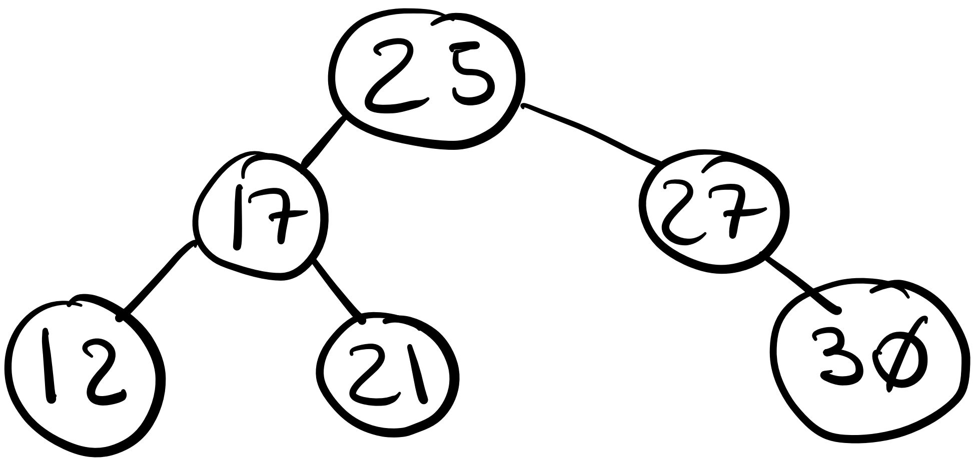


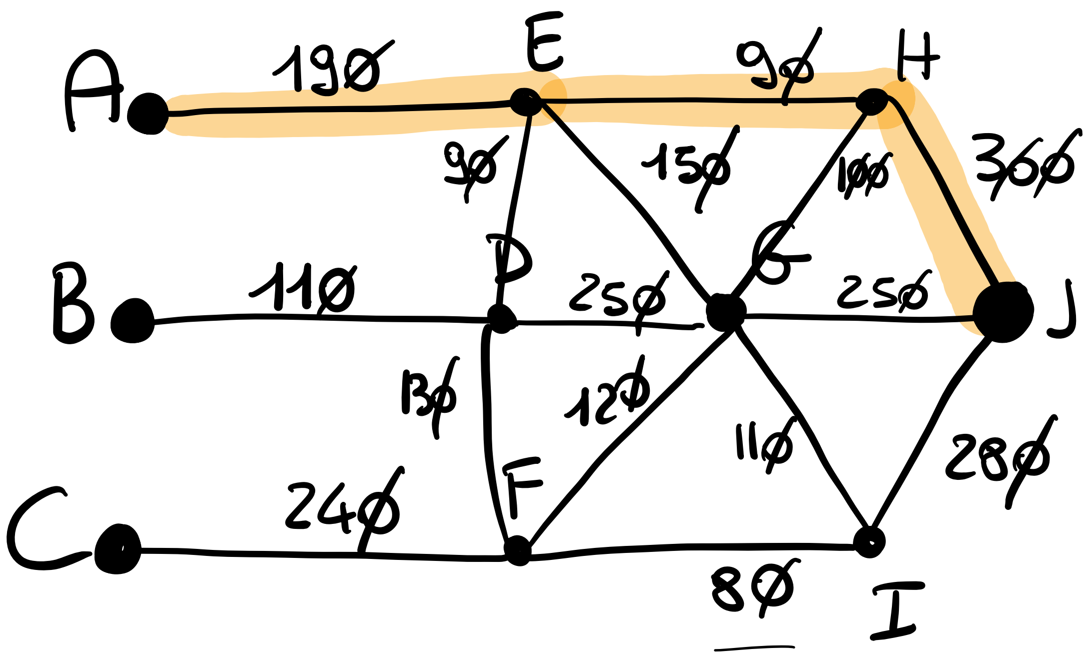


21
17
DS
12 -2
6
17 -1
15
21 -1
25
DD
17
12
6
15
21 -1 -1 = -2
25 -1
27
DD



$\boxed{\text{DD}}$

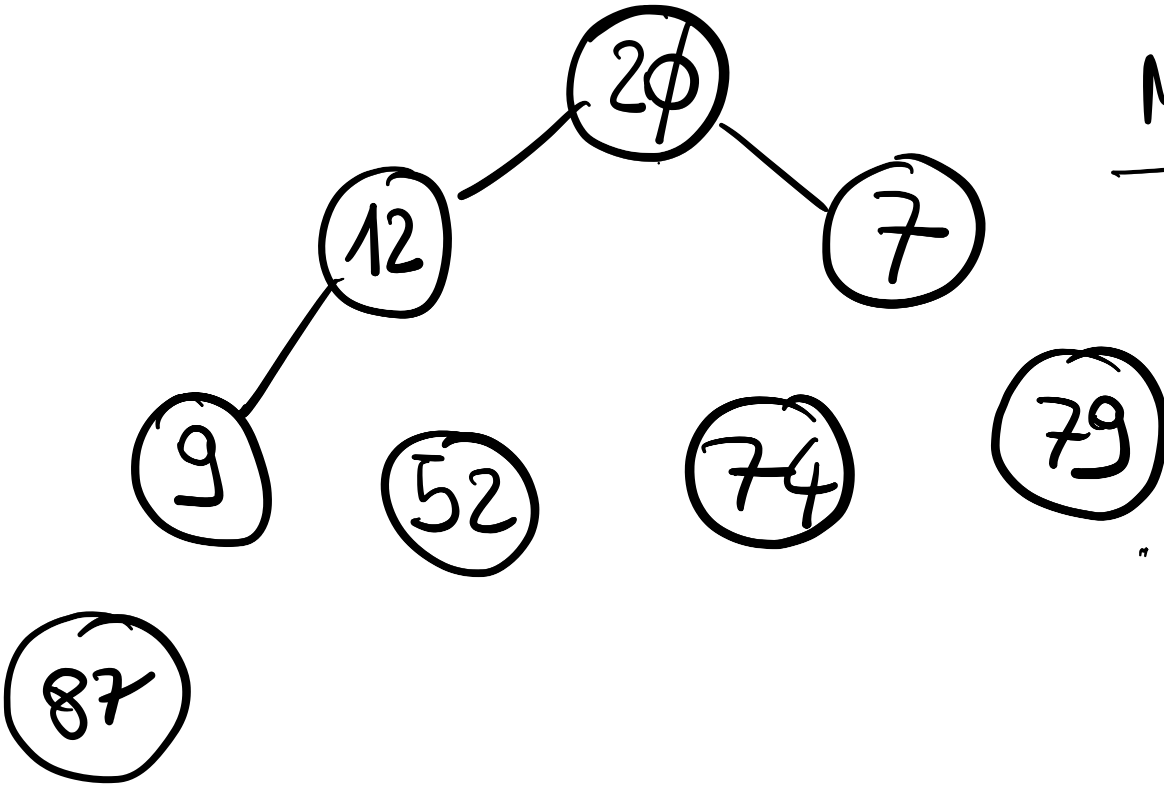


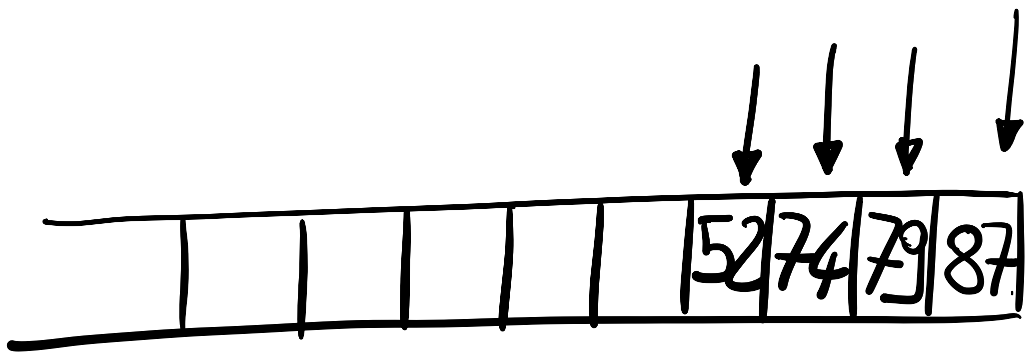


	A	B	C	D	E	F	G	H	I	J
	∞ N	∞ N	∞ N	∞ N	∞ N	∞ N	∞ N	∞ N	∞ N	Ø N
	∞ N	—	∞ N	∞ N	∞ N	∞ N	250, J	300, J	280, I	
	∞, N	∞, N	∞, N	500, G	400, G	370, G		300, J	280, I	
	∞ N	∞ N	∞ N	500, G	400, G	360, I		300, J		
	∞ N	∞ N	∞ N	500, G	390, H	360, I				
	∞ N	∞ N	600, F	430, F	390, H					
	580, E	∞ N	600, F	480, E						
	580, E	590, D	600, F							

79, 87, 7, 9, 12, 74, 52, 2~~0~~

MAX-HEAP





T $m = 11$

16, 30, 38, 26, 41, 45, 37, 23, 32

1) $(K + i) \bmod 11 \longrightarrow K \bmod 11$

2) $(K \bmod 11 + i \cdot (K \bmod 7 + 1)) \bmod 11$

$i = 0$

T

0	1	2	3	4	5	6	7	8	9	10
	45	23		26	16	38	37	30	41	32

T

0	1	2	3	4	5	6	7	8	9	10
41	45		32	26	16		37	30	38	23

$h(38) = 5 + 38 \bmod 7 + 1 = 9$

$h(41) = 8 + 41 \bmod 7 + 1 = 4$

$8 + 2(41 \bmod 7 + 1) = 0$

$h(37) = 4 + (37 \bmod 7 + 1) = 7$

$h(23) = 1 + (23 \bmod 7 + 1) = 4$

$$h(23) = 1 + 2(23 \bmod 7 + 1) = 7$$

$$h(23) = 1 + 3(\ 2 + 1) = 10$$

$$h(32) = 10 + (32 \bmod 7 + 1) = 4$$

$$h(32) = 10 + 2 \cdot 5 = 9$$

$$h(32) = 10 + 3 \cdot 5 =$$

$$T(n) = 2T(n/3) + (\log n)^2$$

$$n^{\log_3 2} \quad // \quad (\log n)^2$$

$$\sim n^{0.6} \qquad (\log n)^2 \qquad \boxed{\text{CASO 1}}$$

$$\Theta\left(n^{\log_3 2}\right)$$

/* N intero positivo > 0 */

i = 0 ; j = 0 ; k = 0

for i = 0 to N $\longrightarrow O(N)$

{ $\boxed{j = 1}$

  while j < N $\longrightarrow$ 1, 2, 4, 8, 16 ..... $2^i < N \Rightarrow \log_2(N)$

  { $\boxed{k = 1}$

    while k < N $\longrightarrow$ 1, 3, 9, 27 .... $3^i < N \Rightarrow \log_3(N)$

    { i = i + 1

      $\boxed{k = k \cdot 3}$

    }

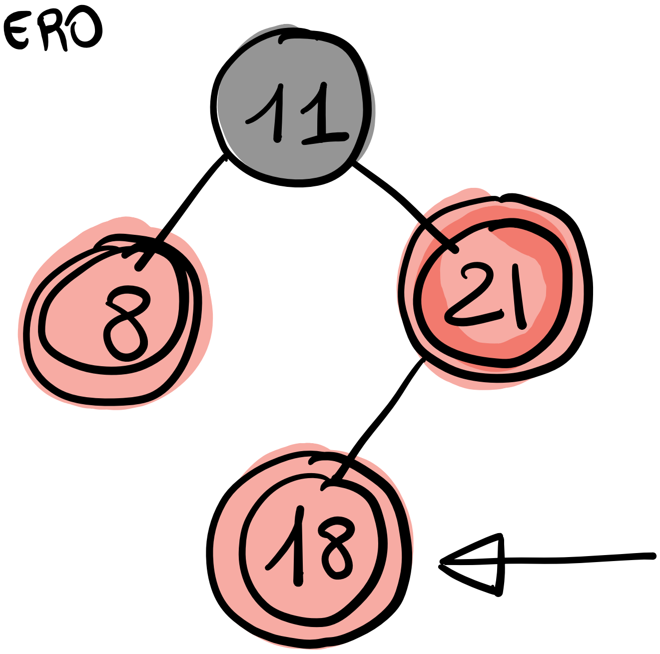
    $\boxed{j = j \cdot 2}$

}

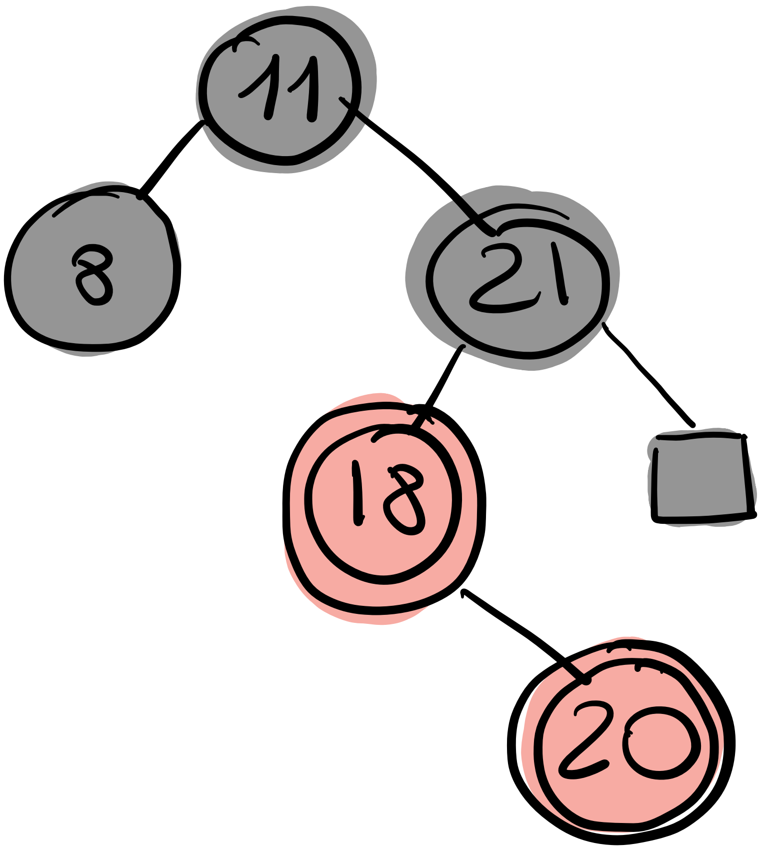
$$O\left(N \cdot (\log N)^2\right)$$

11, 21, 8, 18, 2~~0~~, 28, 43, 83

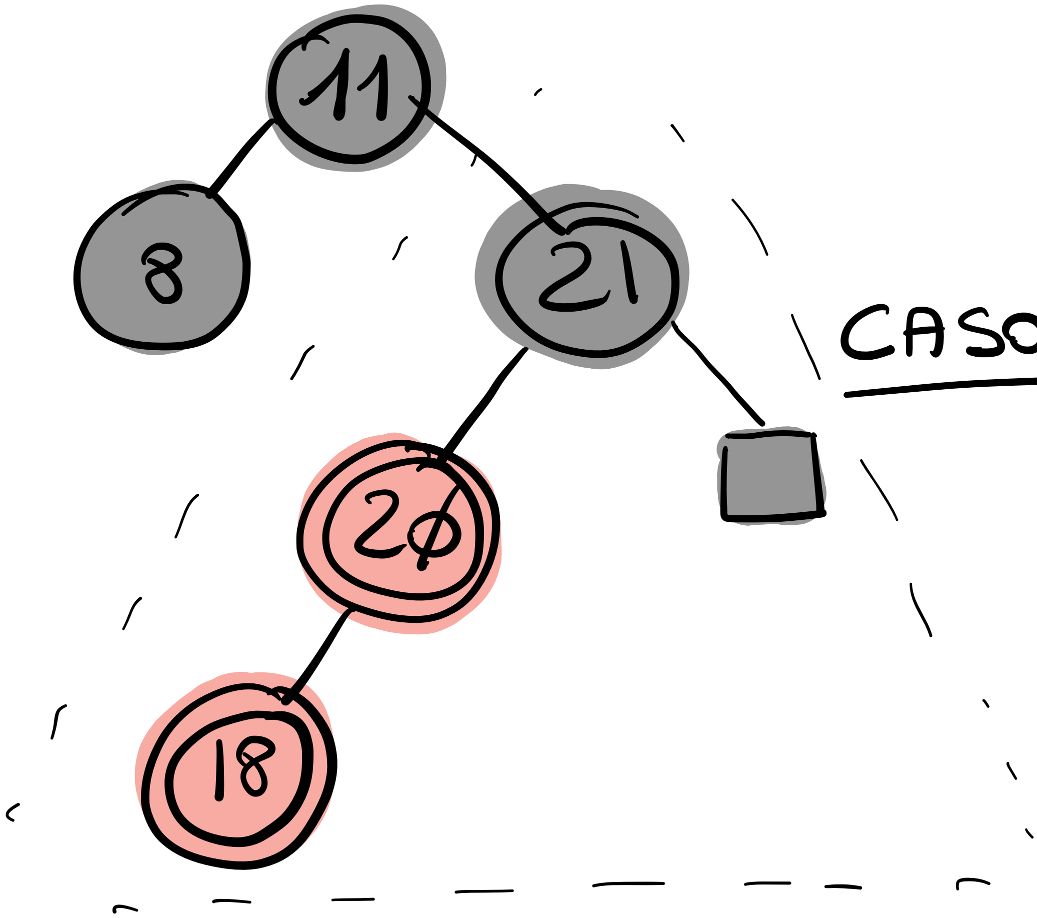
ROSSO-NERO



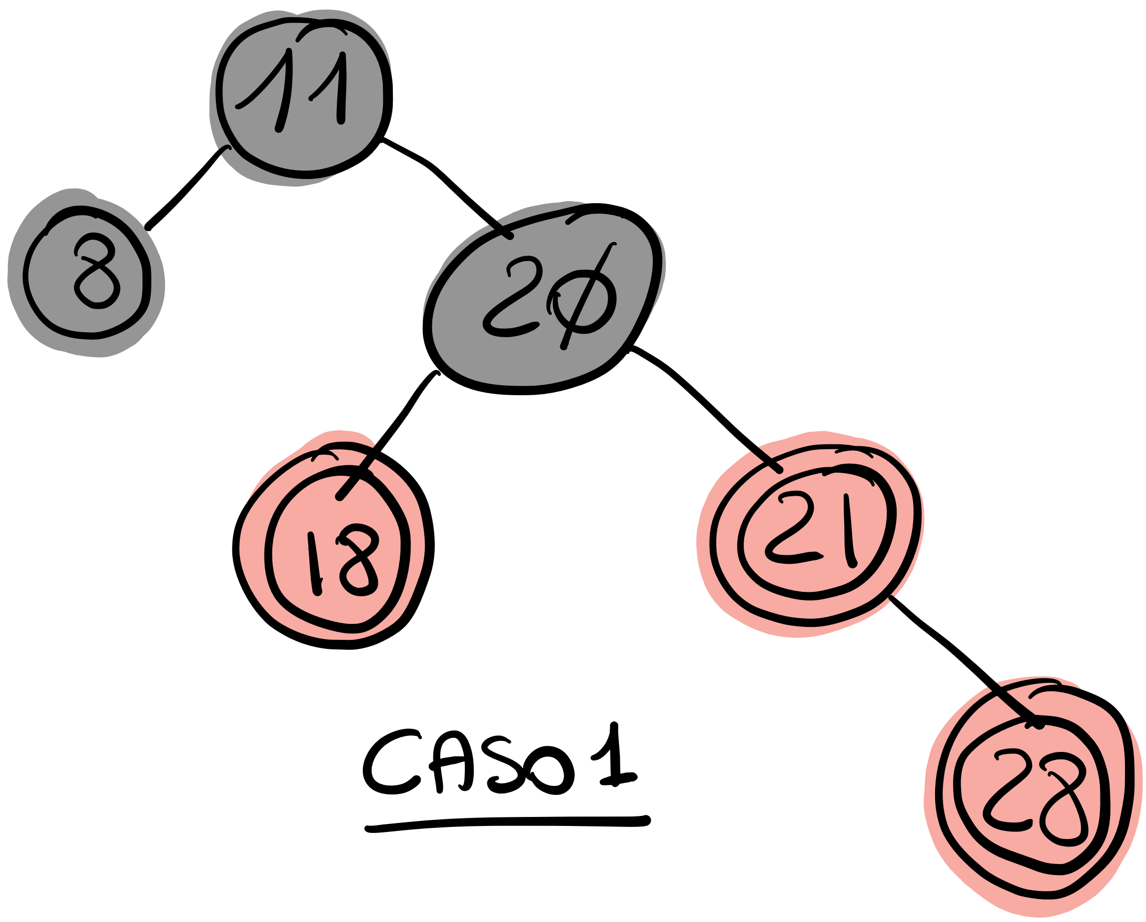
CASO 1



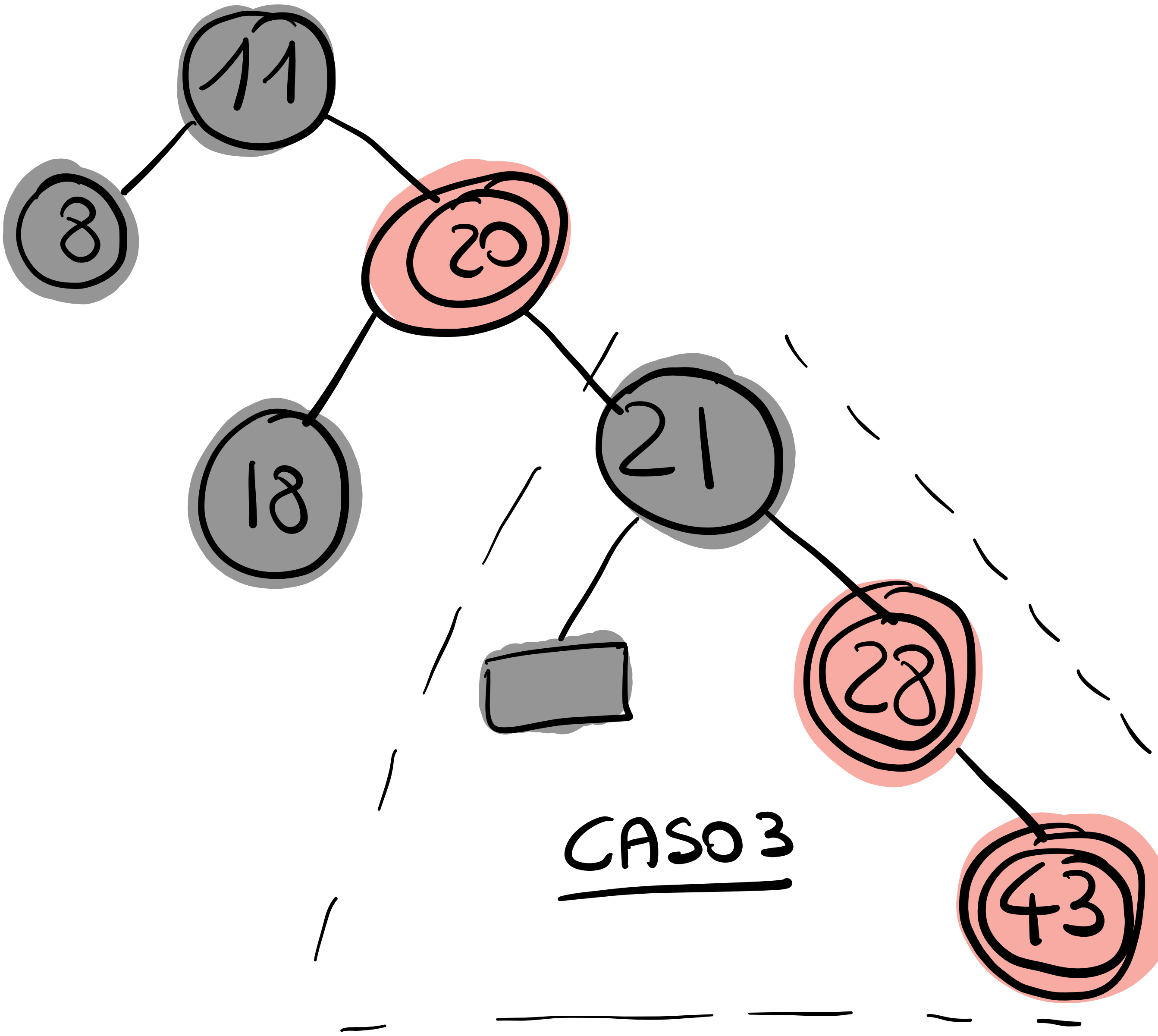
CASO 2

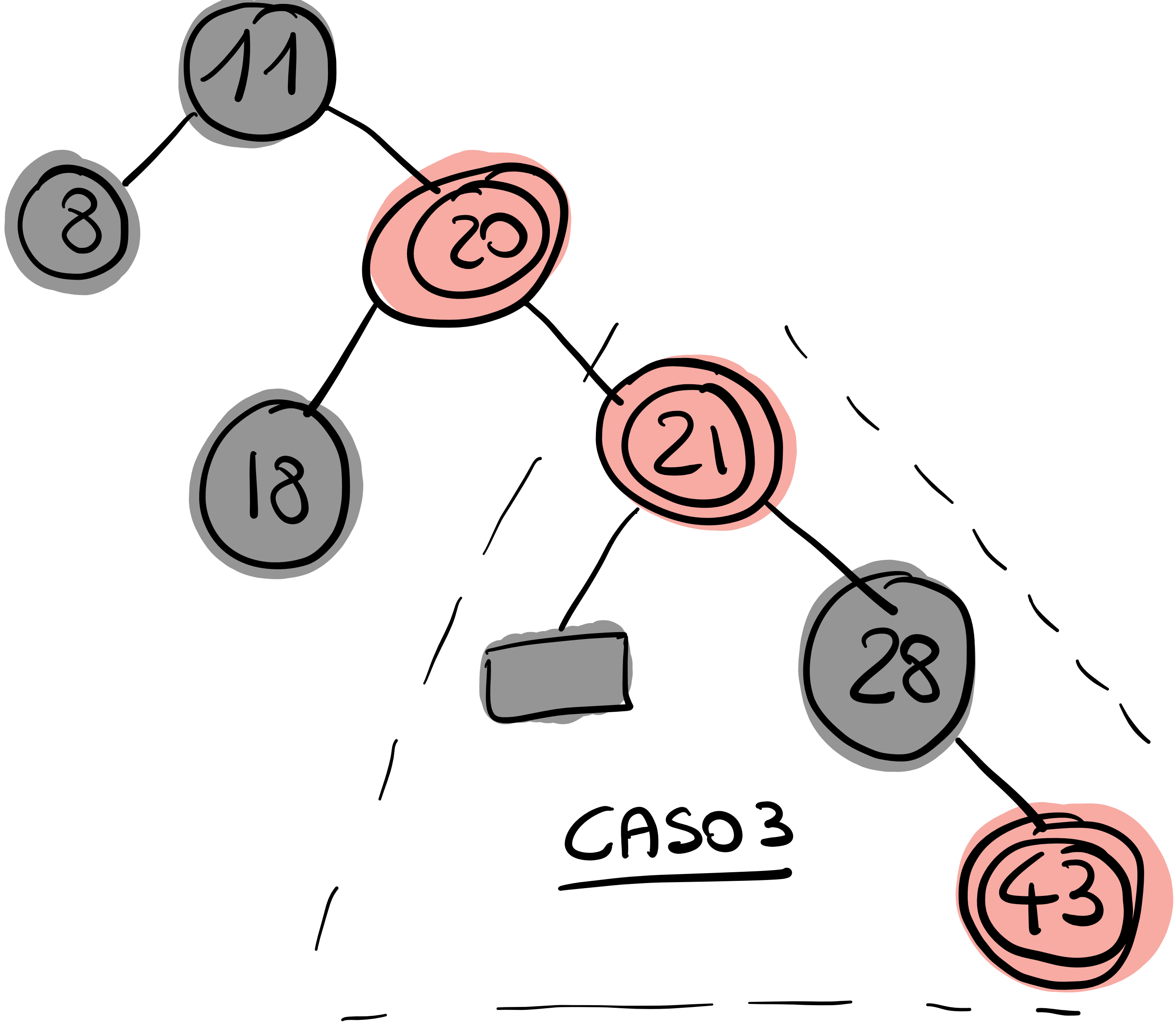


CASO 3

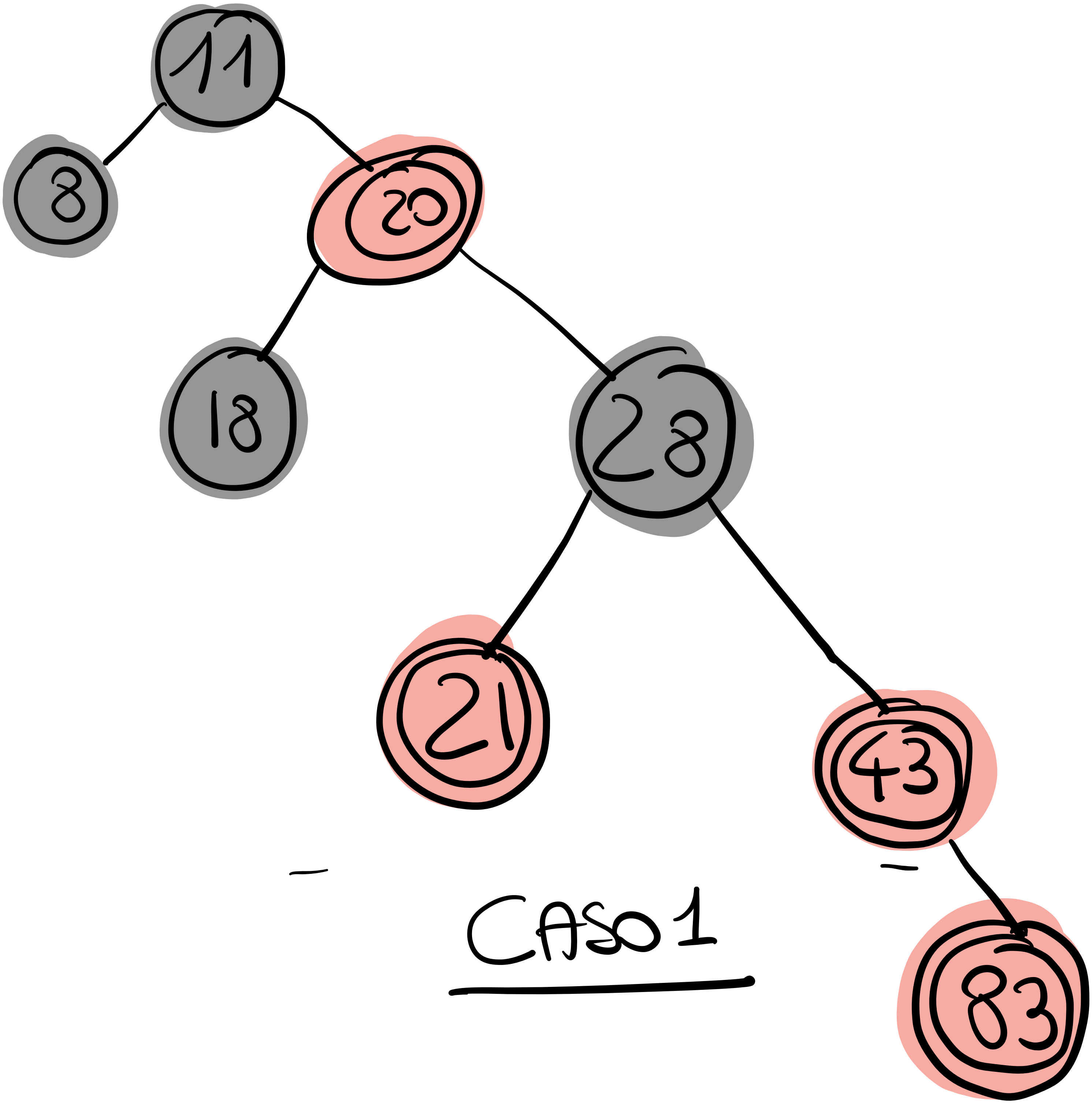


CASO 1


11
8
20
18
21
28
43
CASO 3

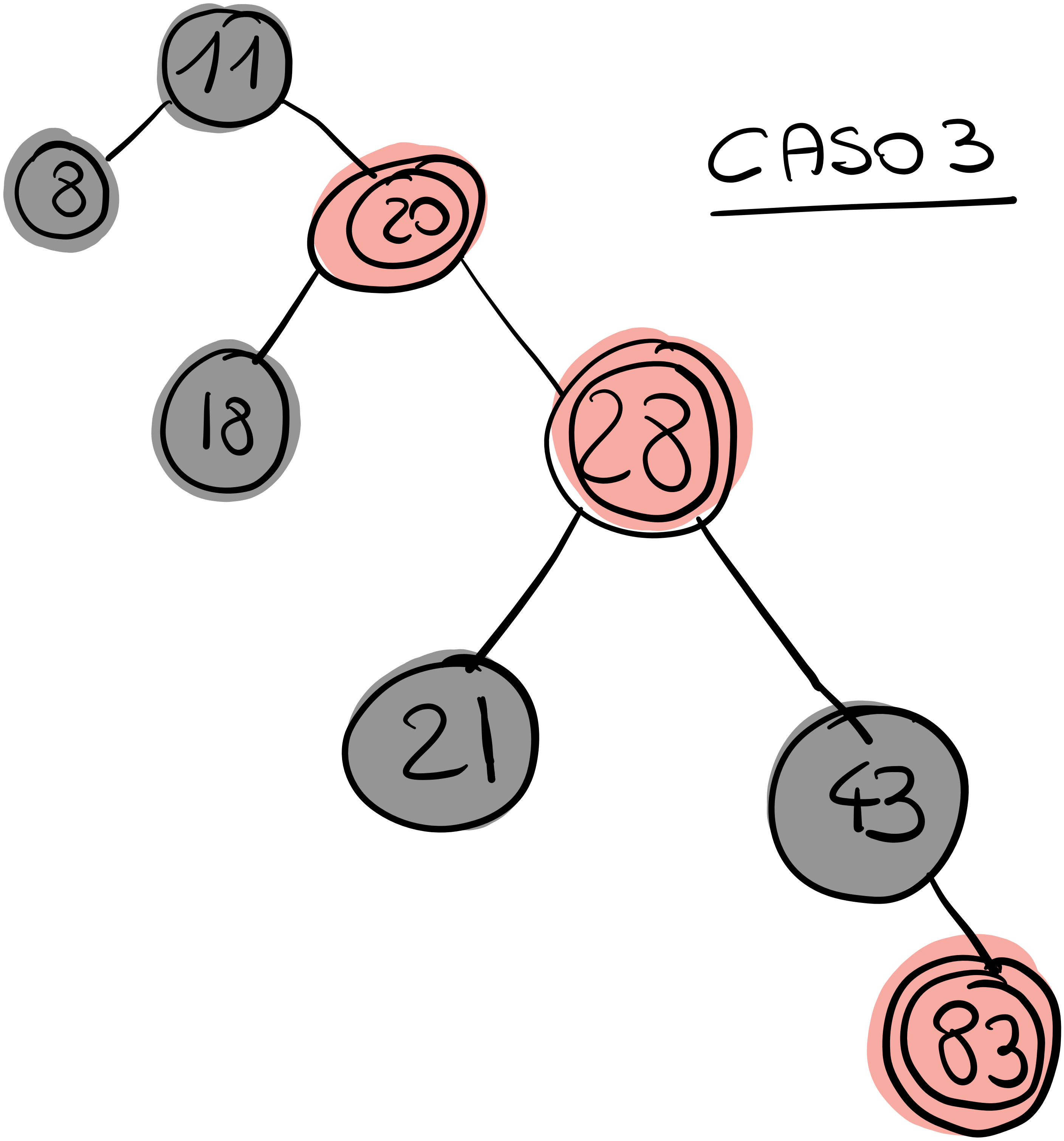


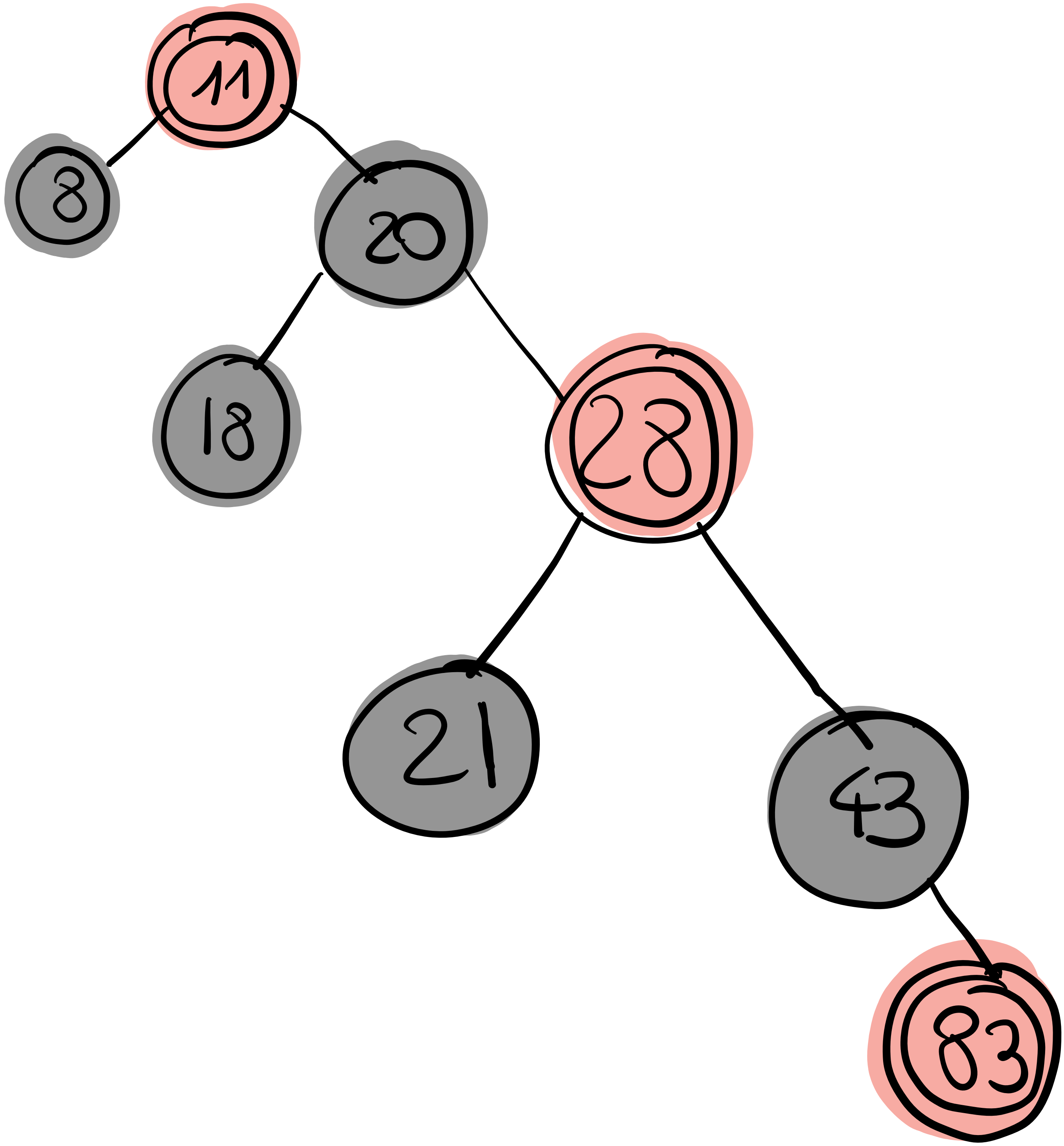
11
8
20
18
21
28
43
CASO 3

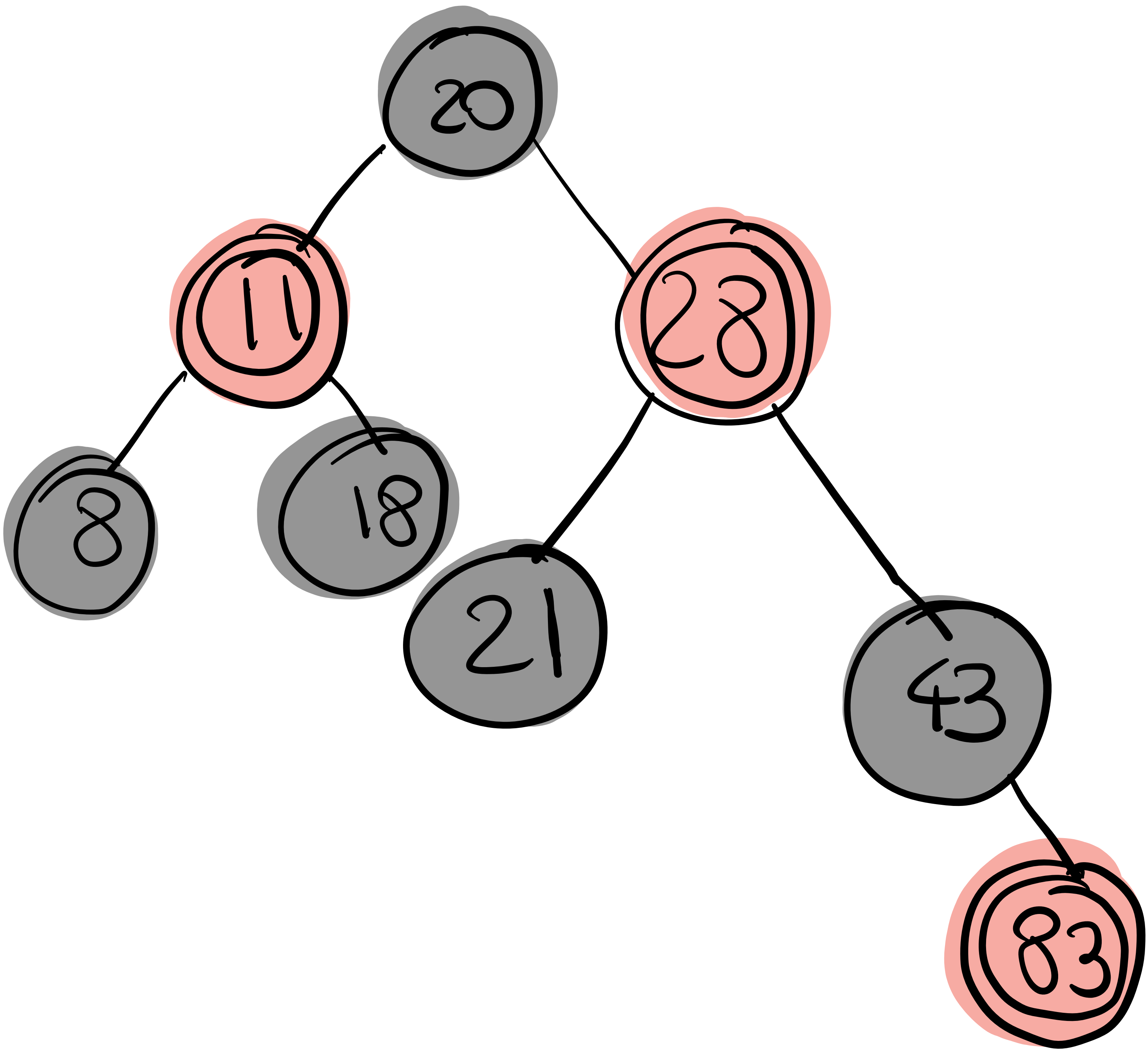


CASO 1

CASO 3




11
8
20
18
28
21
43
83


20
11
28
8
18
21
43
83